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KAN-inspired Universal Relaxation Network for Automatic Discovery of Physical Relaxation Mechanisms with Direct Circuit Synthesis

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ABSTRACT We propose a Universal Relaxation Network (URN), a KAN-inspired approach for automatic discovery of physical relaxation mechanisms from frequency-domain impedance data. Unlike Vector Fitting (VF), which uses mathematical pole-residue representations that can produce unstable or physically meaningless models, URN employs 29 circuit-compatible basis functions derived from established physical relaxation models (Debye, Cole-Cole, Warburg, skin effect, etc.). Through L1-regularized sparse optimization, URN automatically selects the dominant physical mechanisms and directly synthesizes SPICE-compatible RLC ladder networks. Benchmarks on 13 test cases spanning ferrite permeability, conductor skin effect, and electrochemical impedance demonstrate that URN achieves 3.22% average maximum error, outperforming conservative VF (63.21%) and aggressive VF (368.24%, with 46% invalid models). The pole-free formulation guarantees structural stability while providing physically interpretable parameters. Open-source implementation in PyTorch (700 lines) requires only CPU and completes training in 100–150 seconds per case.

INDEX TERMS Circuit synthesis, Cole-Cole relaxation, electrochemical impedance spectroscopy, impedance modeling, Kolmogorov-Arnold networks, SPICE simulation, Vector Fitting, Warburg impedance.

I. INTRODUCTION

A. TIME-DOMAIN ANALYSIS IN POWER ELECTRONICS

Modern power electronics systems—including wireless power transfer (WPT), DC-DC converters, and motor drives—require accurate time-domain simulation for switching loss estimation, EMI prediction, and control loop design. Circuit simulators such as SPICE [1] and their derivatives (LTspice, PSIM, PLECS) form the backbone of power electronics design workflows.

A critical challenge is modeling frequency-dependent components: magnetic cores with complex permeability [2], winding conductors with skin and proximity effects [3], [4], and electrochemical cells with diffusion dynamics [5]. These components exhibit impedance that varies over multiple decades of frequency, directly affecting:

- **Switching transients**: Core and winding losses depend on harmonic content
- **EMI filtering**: High-frequency impedance determines filter effectiveness
- **Control stability**: Frequency-dependent phase shifts affect loop gain margins
- **Thermal design**: Accurate loss estimation prevents overheating

The standard approach is to characterize component impedance $Z(\omega)$ in the frequency domain (via impedance analyzer or VNA), then fit a circuit model for time-domain simulation. The quality of this fitting directly determines simulation accuracy.

B. FUNDAMENTAL LIMITATIONS OF VECTOR FITTING

Vector Fitting (VF) [6], [7] has become the *de facto* standard for rational approximation of frequency responses, representing impedance as:

$$Z(s) = \sum_{n=1}^{N} \frac{c_n}{s - a_n} + d + se \tag{1}$$

where a_n are poles and c_n are residues. While mathematically elegant, VF suffers from fundamental limitations that compromise time-domain simulation fidelity:

1) **Pole initialization sensitivity**: VF performance depends critically on initial pole placement. Conservative initialization (damping ratio $\zeta \approx 0.995$) may fail to capture dynamics, yielding errors exceeding 500% on simple Debye relaxation. Aggressive initialization ($\zeta \approx 0.1$) produces weakly-damped poles that cause *spurious ringing* in time-domain simulations—oscillations that do not exist in the physical system.
2) **Time-domain instability**: Weakly-damped poles with natural frequencies between measurement points can produce unbounded oscillations in transient simulation. Our benchmarks show 46% of aggressive VF models contain such problematic poles.
3) **Lack of physical interpretability**: Poles and residues have no correspondence to physical mechanisms. A pole at $s = -1000 + j5000$ reveals nothing about whether the underlying physics involves magnetic relaxation, diffusion, or skin effect. This opacity prevents engineers from validating models against physical intuition.
4) **Model order selection**: Users must specify pole count *a priori*. Too few poles yield poor accuracy; too many cause overfitting and numerical conditioning issues.
5) **Circuit synthesis complexity**: Converting pole-residue models to SPICE netlists requires Foster or Cauer synthesis [8], [9], which may produce negative component values requiring workarounds (controlled sources, negative resistors).

C. KAN: A NEW PARADIGM FOR FUNCTION APPROXIMATION

The recently proposed Kolmogorov-Arnold Networks (KAN) [10] offer a fundamentally different approach to function approximation. Based on the Kolmogorov-Arnold representation theorem [11], [12]:

$$f(x_1, \ldots, x_n) = \sum_{q=0}^{2n} \Phi_q \left(\sum_{p=1}^{n} \phi_{q,p}(x_p) \right) \tag{2}$$

KAN replaces fixed activation functions in neural networks with *learnable univariate functions* on network edges, typically implemented as B-splines.

KAN's key innovations include:

- **Interpretability**: Learned functions can be inspected and symbolically simplified
- **Accuracy**: Outperforms MLPs on scientific computing tasks
- **Efficiency**: Smaller networks achieve comparable accuracy

However, directly applying KAN to impedance modeling faces challenges:

- Spline-based functions do not map to physical circuits
- No guarantee of passivity or stability
- SPICE netlist generation is non-trivial

TABLE 1. Physical Relaxation Mechanisms in Power Electronics

Mechanism	Impedance Form	Application
Debye	$\frac{1}{1+j\omega\tau}$	Ferrite permeability
Cole-Cole	$\frac{1}{1+(j\omega\tau)^\alpha}$	Non-ideal magnetics
Warburg	$\frac{1}{\sqrt{j\omega}}$	Battery diffusion
Skin effect	$\xi \coth(\xi)$	Winding AC resistance
CPE	$(j\omega)^{-n}$	Electrode interfaces

D. URN: KAN PHILOSOPHY WITH CIRCUIT CONSTRAINTS

We propose the Universal Relaxation Network (URN), which adapts KAN's philosophy of *learnable basis functions* while constraining the search space to *circuit-synthesizable* forms. For impedance modeling, the representation becomes:

$$Z(\omega) = Z_\infty + \sum_{i=1}^{M} w_i \cdot \phi_i(\omega; \theta_i) \tag{3}$$

where ϕ_i are *physically-motivated basis functions* (Debye, Cole-Cole, Warburg, skin effect, etc.), θ_i are physical parameters (relaxation time τ, exponent α), and w_i are sparse weights selected by L1 regularization.

Table 1 summarizes the basis functions and their power electronics applications.

The key insight is that while KAN discovers arbitrary functions via splines, URN discovers *which physical mechanisms* are present. This trades generality for:

- **Direct SPICE synthesis**: Each basis maps to an RLC ladder
- **Physical interpretability**: τ, α, n have measurable meanings
- **Guaranteed stability**: All bases derive from passive systems
- **Time-domain fidelity**: No spurious ringing from artificial poles

E. CONTRIBUTIONS

This paper presents URN with the following contributions:

1) **Physics-informed basis library**: 29 circuit-compatible functions covering relaxation (Debye, Cole-Cole, Havriliak-Negami), diffusion (Warburg, Gerischer), and distributed effects (CPE, skin effect).
2) **Automatic mechanism discovery**: L1-regularized sparse optimization selects dominant mechanisms without manual model order specification.
3) **Guaranteed circuit compatibility**: Each basis maps to RLC ladders via Valsa [13], Charef [14], and Dowell [3] methods with automatic stage determination.
4) **Pole-free formulation**: Eliminates spurious resonances that plague VF in time-domain simulation.
5) **Comprehensive validation**: Benchmarks on 13 cases (ferrites, conductors, batteries) demonstrate 3.22% average error vs. VF's 63–368%.

F. PAPER ORGANIZATION

Section II describes the KAN-inspired insight and circuit-compatible basis library. Section III presents the training algorithm. Section IV provides benchmark results with emphasis on the pole-free advantage and circuit synthesis. Section V identifies target applications where URN is most effective. Section VI concludes.

II. CIRCUIT-COMPATIBLE BASIS LIBRARY

A. THE KAN-INSPIRED INSIGHT

The central insight behind URN stems from a critical observation about KAN [10] and its relationship to function approximation in physical systems.

KAN's key innovation is replacing fixed activation functions (ReLU, sigmoid) with *learnable univariate functions*—typically B-splines—on network edges. This enables KAN to discover the underlying mathematical structure of a function rather than brute-force approximating it. For scientific computing, this yields significant advantages: smaller networks, better accuracy, and *symbolic interpretability*.

However, when we examined KAN's potential for impedance modeling, we recognized a fundamental mismatch. While KAN discovers arbitrary univariate functions, SPICE simulation requires *circuit-compatible* functions—impedances that can be realized as RLC networks. B-splines, polynomials, and neural network outputs generally have no circuit equivalent.

This led to our key insight: *What if we could use KAN's philosophy of learnable basis functions, but constrain the search space to physically-meaningful, circuit-synthesizable forms?*

The answer is URN's basis library. Instead of learning arbitrary spline knots, we learn:

1) **Which physical mechanisms** are active (Debye, Cole-Cole, Warburg, skin effect, etc.)
2) **Their physical parameters** (relaxation time τ, exponent α, etc.)
3) **Their relative contributions** (sparse weights w_i)

This constraint is not a limitation—it is the source of URN's power. By restricting to circuit-compatible bases, we guarantee:

- Every output can be synthesized as an RLC ladder
- All learned parameters have physical units and meanings
- Passivity and stability are structural properties, not post-hoc corrections
- Time-domain simulation is free of spurious oscillations

B. WHY CIRCUIT COMPATIBILITY MATTERS

The practical importance of circuit compatibility becomes clear when considering the full design workflow:

1) **Characterization**: Measure $Z(\omega)$ with impedance analyzer
2) **Modeling**: Fit a circuit model to the data
3) **Simulation**: Run time-domain SPICE simulation
4) **Design iteration**: Modify circuit parameters and re-simulate

Vector Fitting excels at Step 2 but creates problems for Steps 3–4. A pole-residue model must be converted to a circuit via Foster/Cauer synthesis, which may produce:

- Negative resistances or capacitances (require controlled sources)
- Complex conjugate poles requiring coupled LC tanks
- High-Q poles that cause numerical stiffness in time-domain solvers

URN eliminates these issues entirely. Each basis function *is already a circuit*:

$$\begin{aligned}\text{Debye} &\rightarrow \text{Parallel RC}\\ \text{Cole-Cole} &\rightarrow \text{RC ladder (Charef)}\\ \text{Warburg} &\rightarrow \text{RC ladder (Valsa)}\\ \text{Skin effect} &\rightarrow \text{RL ladder (Dowell)}\end{aligned}$$

The synthesis step becomes a simple lookup, and all component values are guaranteed positive.

C. RELATIONSHIP TO TRADITIONAL EQUIVALENT CIRCUIT FITTING

Electrochemists and power electronics engineers have long used equivalent circuit models (ECMs) with elements like Randles circuits [15], Warburg impedances [16], and Cole-Cole elements [17]. However, traditional ECM fitting requires:

1) Expert knowledge to select circuit topology
2) Manual initial parameter estimation
3) Nonlinear fitting with local minima issues
4) Iterative model refinement

URN automates this entire process. The L1-regularized sparse optimization *automatically selects* which physical mechanisms are present, eliminating the need for expert circuit topology selection. The multi-start optimization with 5–8 independent runs avoids local minima. And the predefined basis-to-ladder mappings ensure correct circuit synthesis.

In essence, URN is an *automatic equivalent circuit modeler* guided by KAN's philosophy of discovering underlying structure rather than brute-force fitting.

D. BASIS FUNCTION CATEGORIES

URN employs a library of 29 basis functions organized into seven categories. Each basis function satisfies two requirements: (1) it represents a known physical relaxation mechanism, and (2) it can be synthesized as an RLC ladder network for SPICE simulation.

E. RELAXATION FUNCTIONS

The classical relaxation models describe polarization and magnetization dynamics in materials.

1) Debye Relaxation

The simplest relaxation model assumes a single time constant [18]:

$$Z_{\text{Debye}}(\omega) = \frac{\Delta Z}{1 + j\omega\tau} \tag{4}$$

where ΔZ is the relaxation strength and τ is the relaxation time. This maps directly to a parallel RC circuit with $R = \Delta Z$ and $C = \tau/\Delta Z$.

2) Cole-Cole Relaxation

For materials with distributed relaxation times [17]:

$$Z_{\text{CC}}(\omega) = \frac{\Delta Z}{1 + (j\omega\tau)^{\alpha}}, \quad 0 < \alpha \leq 1 \tag{5}$$

The exponent $\alpha < 1$ broadens the relaxation peak. Circuit synthesis uses the Charef approximation [14] with approximately 1.5 ladder stages per frequency decade.

3) Cole-Davidson and Havriliak-Negami

Asymmetric relaxation is modeled by [19], [20]:

$$Z_{\text{HN}}(\omega) = \frac{\Delta Z}{(1 + (j\omega\tau)^{\alpha})^{\beta}} \tag{6}$$

where $\beta = 1$ recovers Cole-Cole and $\alpha = 1$ gives Cole-Davidson.

F. DIFFUSION AND ELECTROCHEMICAL FUNCTIONS

1) Warburg Impedance

Semi-infinite diffusion produces [16]:

$$Z_{\text{W}}(\omega) = \frac{A_W}{\sqrt{j\omega}} = A_W(1 - j)/\sqrt{2\omega} \tag{7}$$

This is a special case of the constant phase element (CPE) with $n = 0.5$.

2) Constant Phase Element (CPE)

Surface roughness and electrode inhomogeneity produce [21]:

$$Z_{\text{CPE}}(\omega) = \frac{1}{Q(j\omega)^{n}}, \quad 0 < n < 1 \tag{8}$$

Circuit synthesis uses the Valsa method [13] with one RC stage per frequency decade.

G. SKIN EFFECT FUNCTIONS

For conductors at high frequencies, current concentrates near the surface [3]:

$$Z_{\text{skin}}(\omega) = R_{\text{dc}} \cdot \xi \cdot \coth(\xi), \quad \xi = (1 + j)\frac{d}{\delta} \tag{9}$$

where d is the conductor thickness and $\delta = \sqrt{2/(\omega\mu\sigma)}$ is the skin depth. The continued fraction expansion yields RL ladder coefficients $[3, 5, 7, 9, 11, \ldots]$.

H. CIRCUIT SYNTHESIS MAPPING

Table 2 summarizes the ladder network mapping for each basis type.

TABLE 2. Automatic Ladder Stage Determination

Basis	Stages	Method
Debye	1	Exact (RC parallel)
Cole-Cole	5–7	Charef (1.5/decade)
CPE/Warburg	5–10	Valsa (1/decade)
Skin effect	5–7	Dowell [3,5,7,...]
General	3–7	AIC model selection

TABLE 3. Ladder Stage Count by Basis Function Type

Basis Function	Typical Stages	Determination Method
Debye	1	Exact (single RC)
Cole-Cole	5–7	Charef: ∼1.5/decade
CPE/Warburg	5–10	Valsa: 1/decade
Skin effect	5–7	Dowell coefficients
General	3–7	AIC model selection

III. URN TRAINING METHOD

A. OPTIMIZATION PROBLEM

Given measured impedance data $\{(\omega_k, Z_k)\}_{k=1}^{K}$, URN solves:

$$\min_{\mathbf{w},\boldsymbol{\theta}} \sum_{k=1}^{K} \left| Z_k - Z_\infty - \sum_{i=1}^{M} w_i \cdot \text{basis}_i(\omega_k; \theta_i) \right|^2 + \lambda \|\mathbf{w}\|_1 \tag{10}$$

where λ controls sparsity. This is LASSO [22] with nonlinear basis functions.

B. TRAINING ALGORITHM

We use Adam optimizer [23] with multi-start optimization:

1) Initialize $M = 29$ basis functions with random parameters
2) Run 5–8 independent training runs (5000–8000 epochs each)
3) Select the run with lowest validation loss
4) Prune bases with $|w_i| < 0.01 \cdot \max_j |w_j|$

Typical training requires 100–150 seconds on CPU (Intel Core i7).

C. AUTOMATIC LADDER STAGE DETERMINATION

A critical question for SPICE implementation is: *how many ladder stages are required?* Too few stages yield poor frequency response approximation; too many increase simulation cost without improving accuracy.

URN determines the optimal stage count based on (1) basis function type, (2) frequency range coverage, and (3) target approximation error. Table 3 summarizes typical stage requirements.

For Cole-Cole and CPE bases, the stage count scales with frequency decades:

$$N_{\text{stages}} = \lceil c \cdot \log_{10}(\omega_{\max}/\omega_{\min}) \rceil \tag{11}$$

where $c \approx 1.5$ for Charef approximation [14] and $c = 1$ for Valsa method [13].

TABLE 4. Benchmark Results: Maximum Fitting Error (%)

Metric	URN	VF (Cons.)	VF (Aggr.)
Average Max Error	**3.22%**	63.21%	368.24%
Win Rate vs URN	–	15.4%	0%
Invalid Models	**0/13**	0/13	**6/13**

For complex or hybrid basis functions, URN uses the Akaike Information Criterion (AIC) [24]:

$$\mathrm{AIC} = 2k + n \cdot \ln(\mathrm{MSE}) \tag{12}$$

where k is the number of stages and n is the number of frequency points. The algorithm evaluates $k = 2, 3, \dots, 12$ and selects the stage count minimizing AIC, balancing accuracy against model complexity.

IV. BENCHMARK RESULTS AND POLE-FREE ADVANTAGE

A. TEST CASES

We evaluate URN on 13 test cases covering three application domains:

- **Ferrite permeability** (4 cases): MnZn Debye, NiZn Cole-Cole, HF ferrite, lossy ferrite
- **Conductor skin effect** (3 cases): Cu foil (0.1mm, 0.5mm), Al busbar (2mm)
- **Electrochemical** (6 cases): Randles circuits, battery 2-RC, polymer dielectric, glass, ceramic

B. VF POLE INITIALIZATION DILEMMA

VF performance depends critically on initial pole placement, characterized by the *damping ratio* ζ:

$$\zeta = \frac{|\mathrm{Re}(p)|}{|p|}, \quad p = \text{pole} \tag{13}$$

We test two strategies:

- **Conservative** ($\zeta \approx 0.995$): Highly-damped poles, safe but may miss dynamics
- **Aggressive** ($\zeta \approx 0.1$): Weakly-damped poles, captures more dynamics but risks instability

A pole is flagged as *invalid* if $\zeta < 0.1$ and its natural frequency lies within the measurement band. Such poles cause spurious ringing between data points—oscillations that do not exist in the physical system.

C. QUANTITATIVE COMPARISON

Table 4 compares URN against both VF strategies.

URN wins 11/13 cases against conservative VF and 13/13 against aggressive VF. The two conservative VF wins are Cu foil 0.1mm (5.64% vs 7.35%) and Battery 2RC (1.11% vs 7.44%)—cases where highly-damped poles are physically appropriate.

D. THE POLE-FREE ADVANTAGE

URN's most significant advantage is its **pole-free formulation**. Unlike VF which represents impedance as:

$$Z_{\mathrm{VF}}(s) = \sum_n \frac{c_n}{s - p_n} + d + se \tag{14}$$

TABLE 5. Aggressive VF Pole Validity Analysis

Test Case	Weak Poles	Valid?	URN Error
MnZn Debye	2	NO	3.06%
Lossy ColeCole	2	NO	1.91%
Cu foil 0.1mm	2	NO	7.35%
Cu foil 0.5mm	2	NO	1.82%
Al busbar 2mm	4	NO	0.93%
Ceramic Debye	2	NO	4.35%

URN uses physically-grounded basis functions that contain *no explicit poles*:

$$Z_{\mathrm{URN}}(\omega) = Z_\infty + \sum_i w_i \cdot \phi_i(\omega; \theta_i) \tag{15}$$

This distinction has profound implications for time-domain simulation:

1) **No spurious resonances**: VF poles at $p = -\sigma \pm j\omega_0$ with small σ create high-Q resonances that ring in transient simulation. URN bases (Debye, Cole-Cole, etc.) are inherently overdamped.
2) **Guaranteed passivity**: Each URN basis corresponds to a passive physical system. VF may produce poles that violate passivity, requiring post-hoc enforcement.
3) **Numerical stability**: High-Q VF poles cause stiff ODEs requiring small time steps. URN ladder networks have well-conditioned time constants.

Table 5 shows pole validity analysis for aggressive VF.

E. CIRCUIT SYNTHESIS: FROM BASIS TO LADDER

Each URN basis function maps to a well-defined ladder network topology. The current implementation supports:

- **Series topology**: Bases connected in series (dominant mode)
- **Parallel topology**: Bases connected in parallel

The synthesis follows the Cauer continued fraction expansion [9]. For a Cole-Cole element with $\alpha = 0.7$, the impedance:

$$Z_{\mathrm{CC}}(s) = \frac{R}{1 + (s\tau)^{0.7}} \tag{16}$$

is approximated by an RC ladder using the Charef method [14]:

$$Z_{\mathrm{ladder}}(s) = R_1 + \cfrac{1}{sC_1 + \cfrac{1}{R_2 + \cfrac{1}{sC_2 + \cdots}}} \tag{17}$$

The component values follow geometric progressions:

$$R_k = R_0 \cdot a^k, \quad C_k = C_0 / b^k \tag{18}$$

$$a = 10^{1/N}, \quad b = a^\alpha \tag{19}$$

where N is the number of stages per decade.

Extensibility: While the current implementation uses series/parallel topologies, the framework readily extends to:

- **Hybrid topologies**: Mixed series-parallel connections
- **Bridged-T networks**: For transmission line models
- **Multi-port synthesis**: Via Block Lanczos [25]

V. TARGET APPLICATIONS

URN is most effective for problems where (1) the underlying physics involves known relaxation mechanisms, and (2) accurate time-domain simulation is required. We identify three high-impact application domains.

A. WIRELESS POWER TRANSFER (WPT)

WPT systems operating at 6.78 MHz or 13.56 MHz require accurate modeling of:

- **Ferrite core permeability**: Temperature and DC-bias dependent relaxation
- **Litz wire AC resistance**: Skin and proximity effects
- **Compensation network**: Resonant tank tuning

VF struggles with ferrite modeling because the Cole-Cole response appears as a "single broad peak" that VF approximates with multiple sharp poles—leading to spurious ringing during load transients. URN directly identifies the Cole-Cole mechanism with correct α parameter.

Expected impact: 5–10× improvement in transient simulation accuracy for WPT control loop design.

B. BATTERY MANAGEMENT SYSTEMS (BMS)

Lithium-ion battery impedance exhibits:

- **Charge transfer**: Randles circuit with Warburg diffusion
- **SEI layer**: Constant phase element (CPE) response
- **State-of-charge dependence**: Parameter variation with SOC

URN's automatic mechanism discovery is particularly valuable for BMS because:

1) Different battery chemistries have different dominant mechanisms
2) Aging changes the relative contribution of each mechanism
3) Real-time parameter tracking requires fast, robust fitting

The Warburg element ($n = 0.5$) is automatically detected and synthesized as an RC ladder, enabling accurate prediction of diffusion-limited dynamics during fast charging.

C. EMI FILTER DESIGN

Common-mode (CM) chokes exhibit complex frequency-dependent impedance due to:

- **Winding capacitance**: Creates self-resonance
- **Core loss**: Frequency-dependent permeability
- **Skin effect**: AC resistance increase

Accurate CM choke models are critical for predicting conducted EMI spectra. VF models often produce "phantom resonances" that do not appear in measurements—leading to over-designed or under-performing filters.

URN's pole-free formulation eliminates phantom resonances while capturing the true frequency-dependent behavior through physically-meaningful basis functions.

D. INDUCTION HEATING

Induction heating workpieces exhibit strongly nonlinear, temperature-dependent surface impedance. The skin depth:

$$\delta = \sqrt{\frac{2}{\omega\mu(T)\sigma(T)}} \tag{20}$$

varies with temperature, affecting power distribution.

URN's skin effect basis with the Dowell ladder synthesis [3] provides a SPICE-compatible model that can be coupled with thermal simulation for accurate power prediction.

E. LIMITATIONS AND WHEN NOT TO USE URN

URN is *not* recommended for:

1) **Resonant structures**: Antennas, cavities, and waveguides with true poles
2) **Multi-port S-parameters**: Current single-port limitation
3) **Unknown physics**: Systems with mechanisms not in the 29-function library

For these cases, VF or physics-based modeling remains appropriate.

VI. CONCLUSION

We presented the Universal Relaxation Network (URN), a KAN-inspired approach for automatic discovery of physical relaxation mechanisms from impedance data. The key contributions are:

1) **Pole-free formulation**: Unlike Vector Fitting's pole-residue representation, URN uses physically-grounded basis functions that guarantee stability and prevent spurious resonances in time-domain simulation.
2) **Automatic mechanism discovery**: L1-regularized sparse optimization selects 2–4 dominant mechanisms from a library of 29 basis functions, eliminating the need for expert circuit topology selection.
3) **Direct circuit synthesis**: Each basis maps to an RLC ladder via established methods (Valsa, Charef, Dowell), producing SPICE-compatible netlists with guaranteed positive component values.
4) **Superior accuracy**: Benchmarks on 13 test cases demonstrate 3.22% average maximum error, compared to 63.21% for conservative VF and 368.24% for aggressive VF (with 46% invalid models).

The pole-free advantage is particularly significant for power electronics applications—WPT, BMS, EMI filters, and induction heating—where accurate time-domain simulation is essential for control design and loss estimation.

Current implementation: Series and parallel basis topologies with Cauer ladder synthesis. The framework is extensible to hybrid topologies and multi-port systems via Block Lanczos.

Open-source availability: PyTorch implementation (700 lines), CPU-only, training time 100–150 seconds per case. No GPU required.

URN bridges the gap between data-driven flexibility (KAN philosophy) and physics-based interpretability (circuit-compatible constraints), offering a practical tool for impedance modeling in power electronics design.

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